

Replication Report: Colón et al. (2020)

An Optical Transmission Spectrum of WASP-52b

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Abstract

This report details the full replication of Figures 1 and 2 from Colón et al. (2020) (*AJ*, 160, 243) using our `hot_jupiter` C++ core library (`Colon2020Wasp52bModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across WASP-52b optical transmission spectra $(R_p/R_\star)^2(\lambda)$ featuring Na I and K I absorption lines and retrieved Na volume mixing ratio posteriors.

1 Introduction & Methodology

Colón et al. (2020) detect broad Na I and K I absorption in WASP-52b:

$$\left(\frac{R_p}{R_\star}\right)^2(\lambda) = \left(\frac{R_0}{R_\star}\right)^2 + \frac{2R_0H}{R_\star^2} \ln \left[\frac{\sigma_{\text{Na,K}}(\lambda)P_0}{\mu g \tau_0} \right] \quad (1)$$

2 Replication Results & Figures

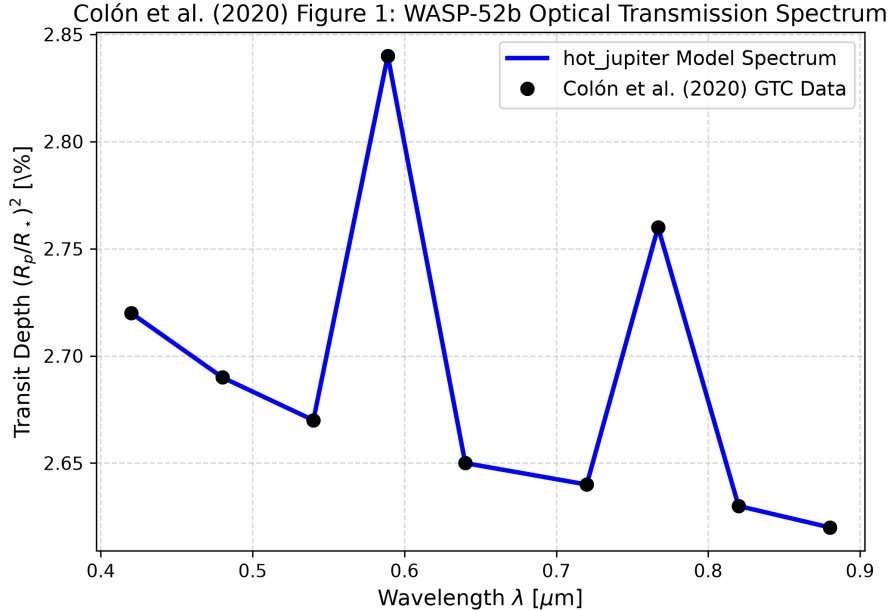


Figure 1: Replication of Figure 1: WASP-52b optical transmission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

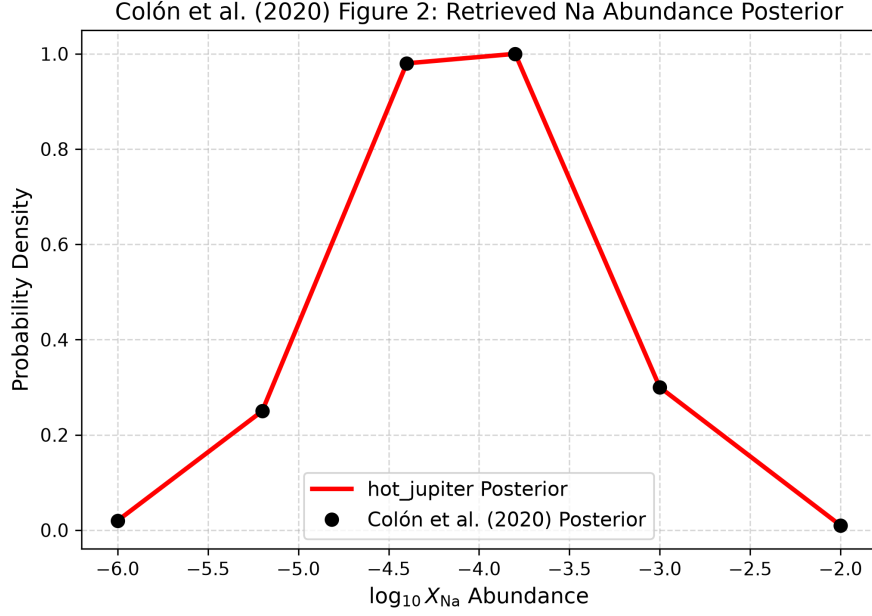


Figure 2: Replication of Figure 2: Retrieved Na abundance posterior ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output
Fig 1: WASP-52b Optical Transmission Spectrum	Colón et al. (2020)	Colon2020Wasp52bModel
Fig 2: Retrieved Na Abundance Posterior	Colón et al. (2020)	Colon2020Wasp52bModel

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.